

# Network Traffic Modelling

440-2216/01 Kvalita služeb • Lecture 04

Jan Rozhon, Miroslav Vozňák • Department of Telecommunications, FEECS

# Where we are going

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Part 1

## Time-domain distributions

Exponential, Weibull, and jitter as Normal

↔ Lecture 02's Poisson process

Part 2

## Markov loss models

Bernoulli → Simple Gilbert → Gilbert →  
Gilbert-Elliott → 4-state

↔ increasing memory, increasing  
parameters

Part 3

## Self-similarity

The Hurst parameter and long-range  
dependence

↔ why "just use Poisson" fails

One question runs through the lecture: **when does the convenient model (memoryless, independent, Gaussian-tailed) stop matching what a real network actually produces?**

# What you should be able to do

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1. Name which quantity each of the exponential, Weibull, and Normal distributions models in network traffic, and why each departs from the simplest available choice.
2. Draw and parametrize the Bernoulli, Simple Gilbert, Gilbert, Gilbert-Elliott, and four-state Markov loss models, and explain what capability each new parameter buys.
3. Define the Hurst parameter and explain what "self-similar" traffic means and why it survives aggregation where a Poisson process does not.
4. Explain why long-range-dependent traffic breaks the delay and buffer-sizing intuition built from Lecture 02's M/M/1 formulas.

Keep asking: **does this model still produce independent, short-memory behaviour — or does it now remember its own past?**

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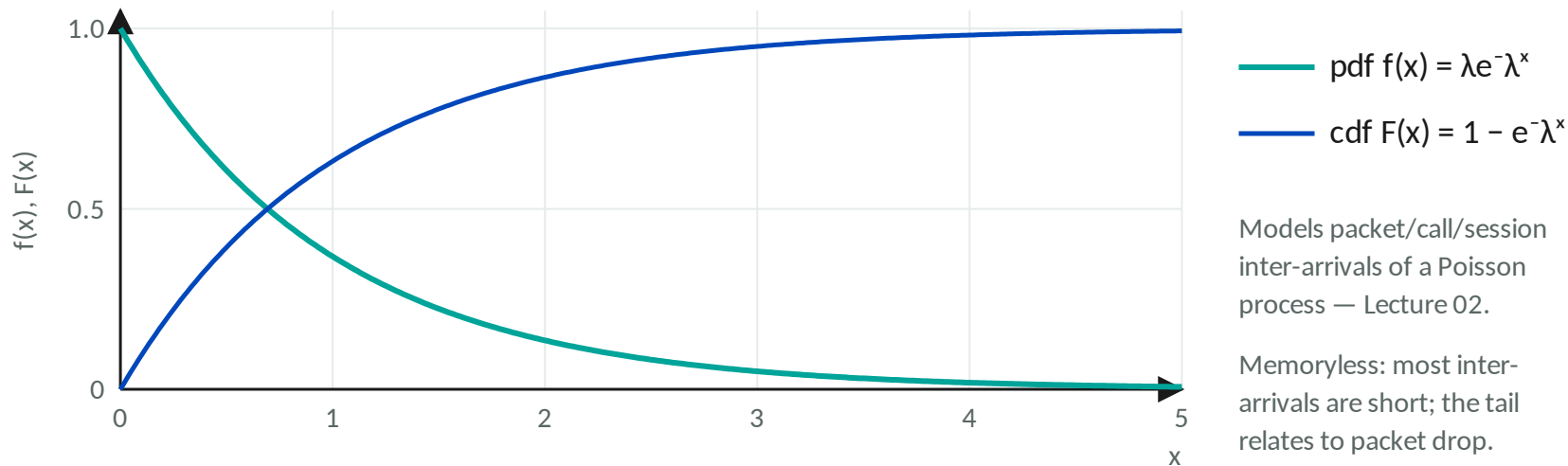
# Real traffic looks random at every time scale you look at it

Modelling it well means choosing the right statistical *structure* — not just the right average

# Part 1

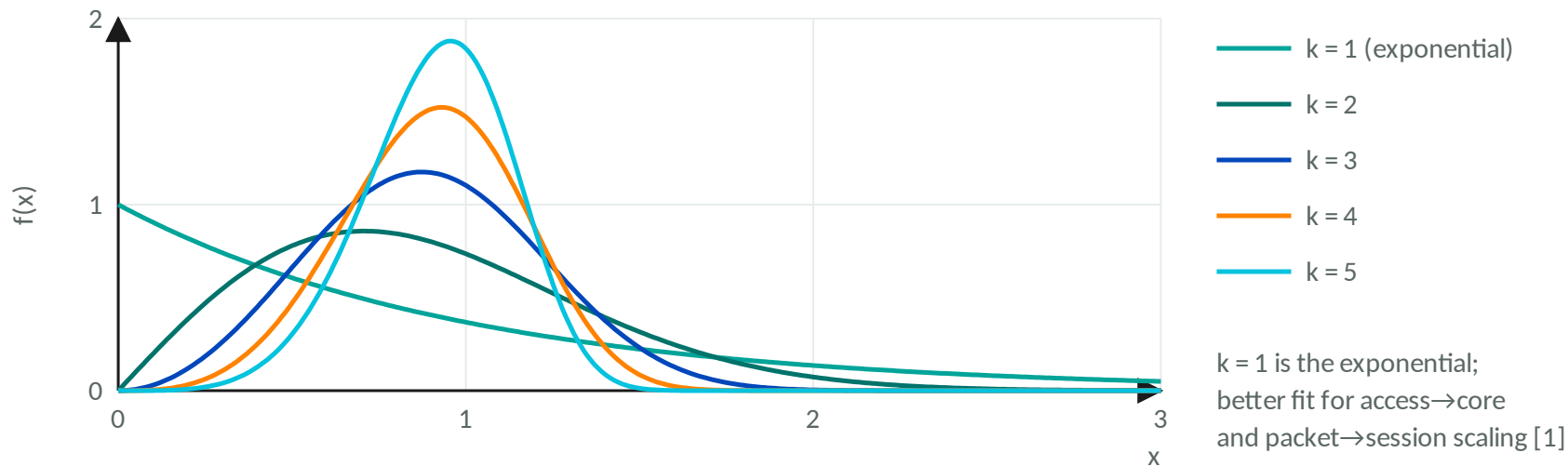
Time-domain traffic distributions

# Exponential distribution — the memoryless baseline



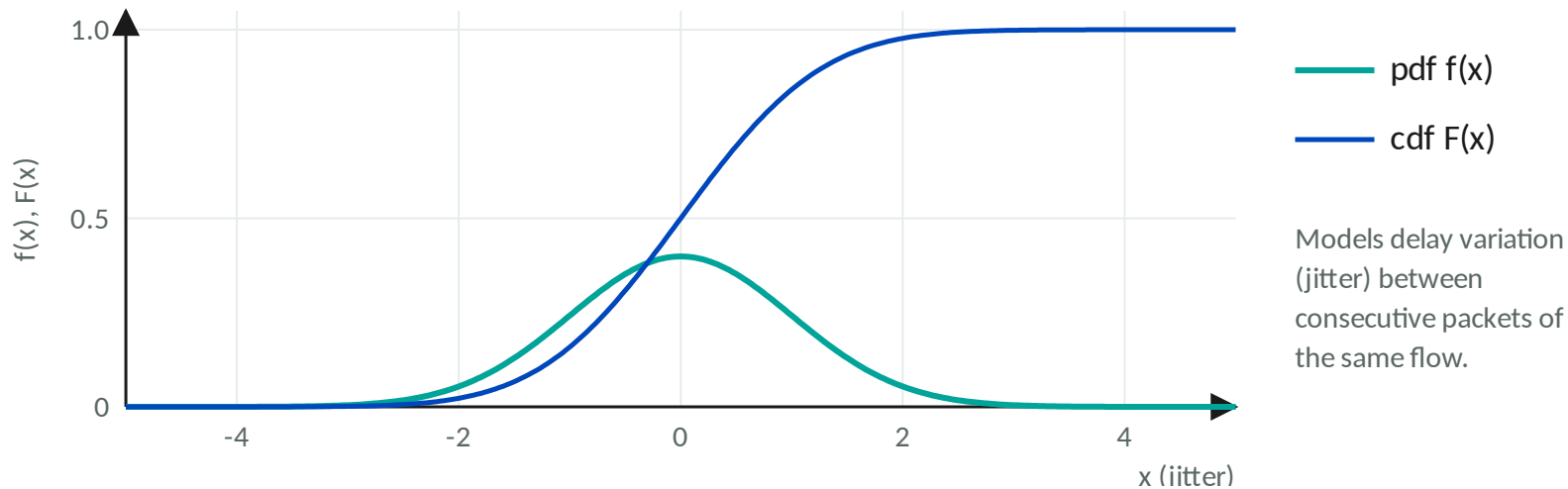
Models packet, call, or session **inter-arrivals** for a Poisson process — exactly Lecture 02's arrival process. Most inter-arrival times are short; the (theoretically infinite) tail is where timeouts and packet drops live.

# Weibull distribution — a better fit at scale



Also models packet, flow, or session inter-arrivals, but with a **shape parameter  $k$**  that the exponential lacks ( $k = 1$  recovers it exactly). Arfeen et al. (2013) found it scales better than the exponential as traffic aggregates from access to core networks, and from packet to session level.

# Normal distribution — jitter



Models the **distribution of packet delay variation (jitter)** for a single flow — not inter-arrival times, but how much each packet's delay deviates from the flow's typical delay. Symmetric around zero, unlike the one-sided exponential and Weibull above.

A playout buffer's size is chosen against this distribution's tail, not its mean.



# Checkpoint — why not just use the exponential everywhere?

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The exponential distribution is simpler than the Weibull and has one fewer parameter to fit. Why does this lecture bother introducing the Weibull at all?

Predict first · discuss with a neighbour · then reveal

**Because measured inter-arrival times often are not memoryless.** The exponential's defining property — the memoryless property,  $P(T > s + t \mid T > s) = P(T > t)$  — is a strong, testable assumption. Real traffic frequently deviates from it, especially as flows aggregate from individual packets up through sessions, and Arfeen et al. found the extra shape parameter  $k$  measurably improves the fit at exactly those aggregation points. Simplicity is not a substitute for matching the data.

# Part 2

Markov-chain packet-loss models

# Markov chains

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Probabilistic models often used to simulate losses in both telco and packet networks — named for **Andrey Markov**, whose first paper on the topic appeared in 1906.

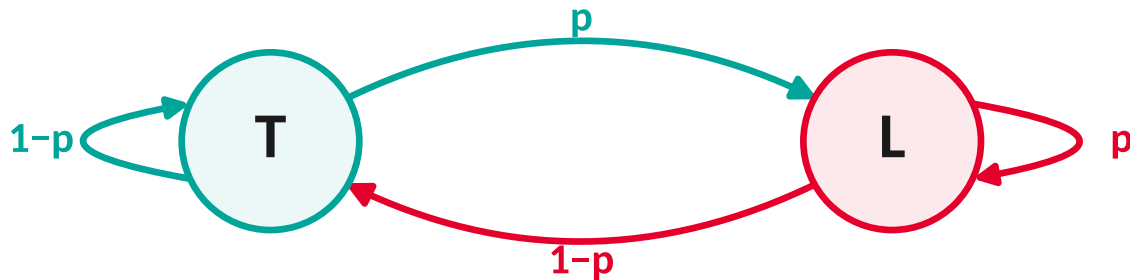
## Features:

- generate a sequence of possible events from a **finite set of states**,
- states are interconnected by **transition probabilities**,
- the current event depends **only on the state attained at the previous event** — no longer memory than that,
- can reach a **steady state**, where the outcome sequence no longer depends on where the model started.

Related tools built on the same idea: **Erlang models** (Lecture 02) and **(Hidden) Markov models**.

# Packet loss — Bernoulli model

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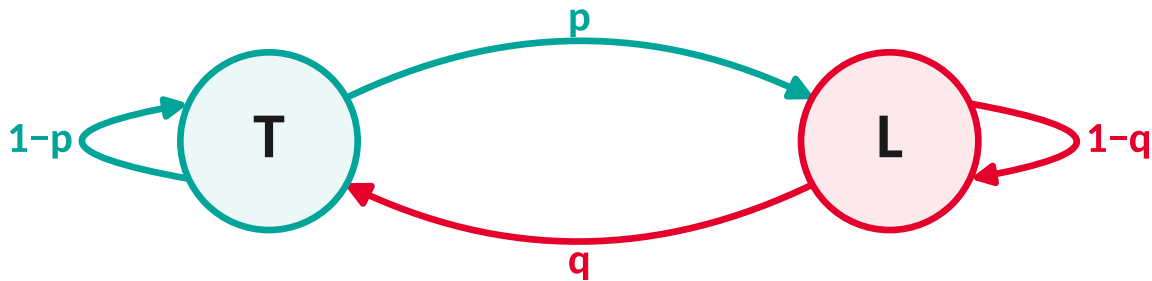


The simplest loss model — a single independent variable  $p$ ; losses are completely independent.

The simplest loss model: just two states and a **single independent variable  $p$** . Every transition into Loss happens with probability  $p$  regardless of the current state — losses are **completely independent** of one another.

# Packet loss — Simple Gilbert model

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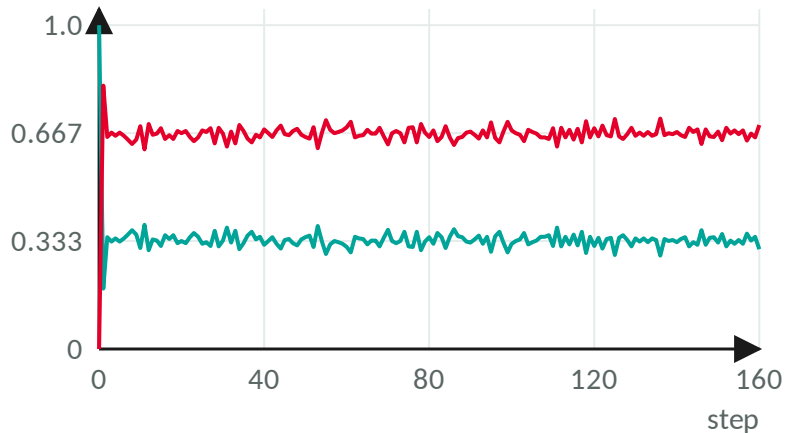


A more elaborate Bernoulli model with a second independent variable  $q$  — can produce loss bursts.

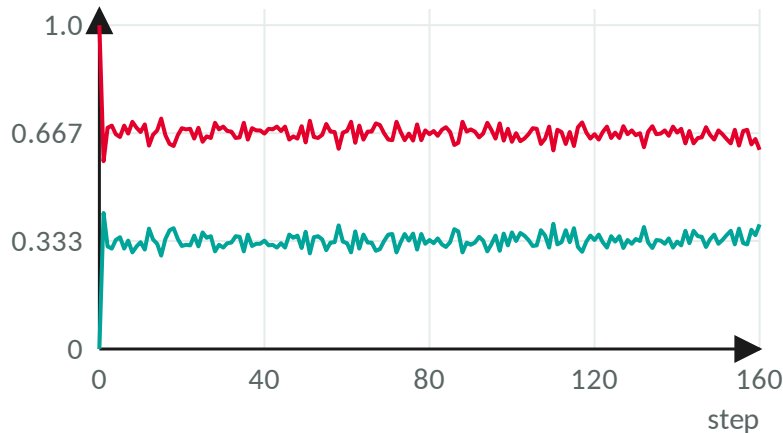
A more elaborate version of the Bernoulli model, adding a **second independent variable  $q$** . Because  $p$  and  $q$  can differ, the chain can now produce **consecutive loss events** that resemble the loss bursts seen in real networks — the Bernoulli model cannot do this at all.

# Simple Gilbert model — convergence to steady state

STARTING IN T



STARTING IN L



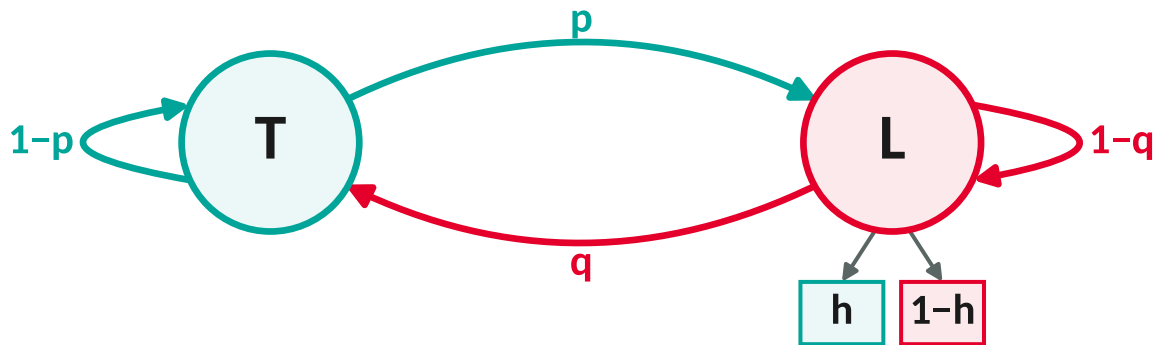
— P(T) — P(L)

$p = 0.8, q = 0.4$  — every realization converges to the same steady state, independent of the start.

With  $p = 0.8, q = 0.4$ , both panels converge to the same steady-state probabilities —  $\pi_L = \frac{p}{p+q} \approx 0.667$  and  $\pi_T = \frac{q}{p+q} \approx 0.333$  — regardless of which state the chain started in.

# Packet loss — Gilbert model

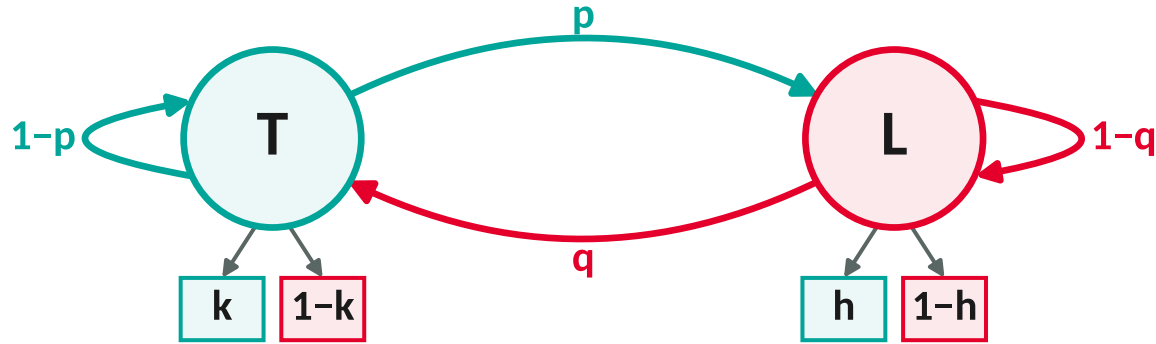
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Even in the "bad" state, the system can still transmit individual packets correctly.

An advancement of the Simple Gilbert model, adding an **emission probability  $h$**  to the "bad" state. Even while the chain is in Loss, the system can still transmit an individual packet correctly — Loss now describes a state of **elevated risk**, not certain loss.

# Packet loss — Gilbert-Elliott model

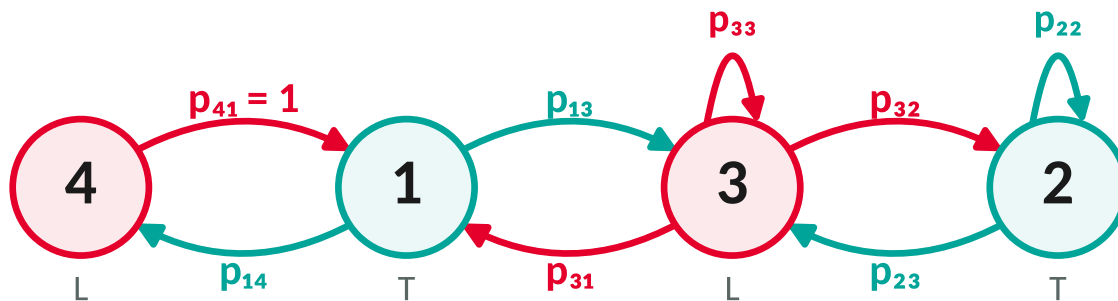


Even in the "good" state, the system can still lose a packet correctly modelled by chance.

An advancement of the Gilbert model, adding a **second emission probability  $k$**  to the "good" state. Even while the chain is in Transmit, the system can still lose a packet — both states now carry their own independent emission probability.



# Packet loss — four-state model



State 4 (isolated loss) always returns to T ( $p_{41} = 1$ ) and never repeats ( $p_{44} = 0$ ); state 3 can repeat itself ( $p_{33}$ ) to model a burst of consecutive losses.

The most advanced model in this group, with **no emission probabilities** at all — the state alone determines whether a packet is lost. Splitting each outcome into two states lets a **short, isolated event** (states 4 and 1) and a **sustained run** (states 3 and 2, which can self-loop) carry independently tunable durations.

## Checkpoint — which model would you reach for?

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You have a measured trace where loss events cluster into bursts of noticeably variable length, and even within a burst, some packets still get through. Which of the five models fits best, and which two capabilities are you actually using?

Predict first · discuss with a neighbour · then reveal

**The Gilbert-Elliott model** is the natural minimum fit: you need state *memory* to produce bursts at all (ruling out Bernoulli), and you need an *emission probability* in the bad state to let some packets through during a burst (ruling out the Simple Gilbert model, which treats Loss as certain loss). Reach for the four-state model only if the burst-length and good-run-length distributions themselves need independent shapes — that model buys duration flexibility, not per-packet risk.

# Part 3

Self-similarity and long-range dependence

# Why the Poisson process isn't enough

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A Poisson process is **bursty on a fine time scale** but **flattens (smooths) on a coarse time scale** — averaged over long enough windows, it looks like white noise. Real network traffic does not do this: it stays visibly bursty however far you zoom out.

**How do you keep burstiness across a wide range of time scales?** Self-similarity.

Features: slow decay of autocorrelation, and **scale-invariance** — the process "looks the same," statistically, whatever time scale it is viewed at.

# Where self-similarity comes from

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The classical explanation (Willinger, Taqqu, Sherman & Wilson, 1997): aggregate traffic is the **superposition of many independent ON/OFF sources** — think individual users or flows, each alternating between sending and idle periods.

**If ON/OFF durations were light-tailed** (e.g. exponential)

the Central Limit Theorem smooths the sum — the aggregate looks Poisson-like at coarse scales, exactly like Lecture 02's arrival process.

**If ON/OFF durations are heavy-tailed** (Pareto, infinite variance)

occasional very long ON or OFF periods keep contributing noticeable structure at every aggregation level — the sum stays self-similar.

# The Hurst parameter

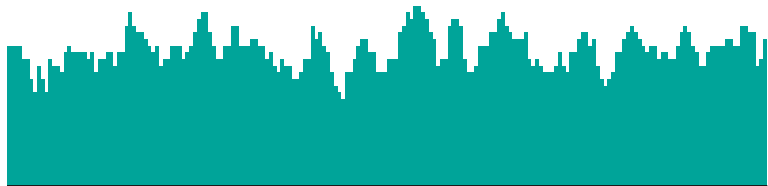
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Introduced by **Harold Hurst (1965)** — a measure of "burstiness," also considered a measure of self-similarity.

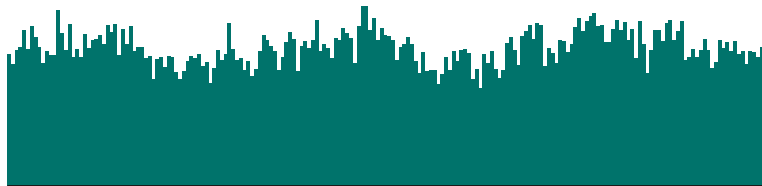
- $0 < H < 1$
- $H$  increases as traffic becomes **more self-similar**
- **white noise** has  $H = 0$
- measures **long-term dependence** of the process
- a parameter of an **infinite series** — for any trace of finite length, it must be **estimated**, not computed exactly
- $H > 0.5$  indicates **long-range dependence (LRD)**: correlations that decay so slowly their sum diverges

# Self-similar traffic across aggregation levels

FINE — 1 bin/bar



×8 aggregation



×64 aggregation



×327 aggregation (whole trace)



A Poisson-based model would flatten towards a smooth average as aggregation grows — this one does not.

A synthetic trace built exactly as the previous slide describes — many ON/OFF sources with Pareto-distributed periods — stays bursty from the finest bin up to the whole trace aggregated into roughly 328-bin blocks.

Leland, Taqqu, Willinger & Wilson's 1994 Bellcore Ethernet measurements are the empirical result that made this phenomenon famous.

# Why this matters for capacity planning

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Lecture 02's queueing formulas —  $W = 1/(\mu - \lambda)$ , buffer occupancy from  $\rho$  — assume **independent, Poisson-like arrivals**. Self-similar, long-range-dependent traffic violates that assumption directly.

Paxson & Floyd's 1995 study, "Wide-Area Traffic: The Failure of Poisson Modeling," is the paper-length version of this point: measured WAN traffic systematically fails Poisson-process tests, and models that ignore this **under-predict** queueing delay and required buffer size.

A link dimensioned from an M/M/1 formula calibrated only to the traffic's **mean rate** will be under-provisioned whenever  $H$  is meaningfully above 0.5 — the sustained above-average periods that self-similarity produces are exactly what an independence-based formula cannot see coming.



## Checkpoint — under-provisioned by how much?

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A link is dimensioned using an M/M/1 formula calibrated to match the traffic's long-run mean rate. The real traffic turns out to be self-similar with  $H \approx 0.8$ . What happens to actual queueing delay relative to the M/M/1 prediction, and why?

Predict first · discuss with a neighbour · then reveal

**Actual delay will run higher than the M/M/1 prediction**, often substantially. M/M/1 assumes independent arrivals, so it only ever "sees" the long-run mean rate. Self-similar traffic with  $H \approx 0.8$  has long-range dependence: sustained periods well above the mean rate are far more likely, and far longer-lived, than an independence-based model allows for — and it is precisely those sustained bursts that build up queueing delay and overflow buffers. The mean rate alone does not determine performance when  $H$  is well above 0.5.

# Common mix-ups

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## Exponential vs. Weibull

Both model inter-arrivals. Exponential is memoryless and a special case ( $k = 1$ ) of Weibull, which fits better once traffic aggregates.

## Emission probability vs. transition probability

$p$  and  $q$  govern the *chain's* state.  $h$  and  $k$  govern whether an *individual packet* is lost, given the current state.

## Steady state vs. self-similarity

A Markov chain's steady state is a long-run *average*. Self-similarity is about variability *persisting* across time scales — the two are unrelated properties.

## $H = 0.5$ vs. $H = 0$

$H = 0.5$  is uncorrelated (Brownian-motion-like) traffic.  $H = 0$  is white noise specifically. Neither is the self-similar regime — that needs  $H > 0.5$ .

## More parameters $\neq$ automatically better

The four-state model is not "more correct" than Gilbert-Elliott — it fits a different feature (independently shaped burst/run lengths) that not every trace needs.

# Summary — three layers of structure

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**Time-domain distributions** Exponential (memoryless,  $k=1$  baseline), Weibull (shape parameter, better fit at scale), Normal (two-sided, models jitter).

**Markov loss models add memory** Bernoulli has none; Simple Gilbert adds a second transition probability; Gilbert and Gilbert-Elliott add emission probabilities for per-packet risk within a state.

**The four-state model separates duration from risk** No emission probabilities — burst length and good-run length each get their own transition structure instead.

**Self-similarity is memory that never fully decays** Long-range dependence ( $H > 0.5$ ) breaks the independence assumption behind Lecture 02's queueing formulas, and under-predicts delay and buffer needs.

## Exit questions — use the ideas

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1. A vendor claims their access-network trace fits an exponential inter-arrival model better than a Weibull one. What would you check before believing that claim?
2. Sketch the state diagram of a loss process where losses are always isolated (never two in a row) but occur with fixed probability  $p$ . Which named model is this?
3. A colleague says "the chain reached steady state, so the traffic must be self-similar." What is wrong with this reasoning?
4. Two traces have the same mean arrival rate; one is Poisson, the other has  $H = 0.85$ . Which one needs a bigger router buffer to hit the same packet-loss target, and why?

**Explain the mechanism**, not just the name.

# Where these models are used

This lecture is a modelling toolbox, not a hands-on exercise on its own — but every piece of it plugs directly into Exercise 02's `simpy`-based pipeline:

- `PacketSource`'s inter-arrival argument accepts **any zero-argument callable**, so an exponential, Weibull, or self-similar generator drops in unchanged.
- A Gilbert/Gilbert-Elliott loss process is a natural model for a lossy `SwitchPort`.

Nothing here requires new tooling — `functools.partial` around a NumPy RNG, exactly as Exercise 02 already uses for inter-arrival and size callables, is enough to try any distribution from Part 1 or 2.

```
1  ssh <lab-server>
2  cd qos-02 && uv sync
3  uv run jupyter lab --ip 0.0.0.0
```

## References — foundations

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- P. A. P. Moran (ed.), A. K. Erlang's original traffic formulas; A. A. Markov, "Rasprostranenie zakona bol'shih chisel na velichiny, zavisyaschie drug ot druga" Extension of the law of large numbers to dependent quantities, 1906.
- M. A. Arfeen, K. Pawlikowski, D. McNickle, A. Willig, "The Role of the Weibull Distribution in Internet Traffic Modeling," Proceedings of the 25th International Teletraffic Congress (ITC), 2013.
- H. E. Hurst, "Long-Term Storage Capacity of Reservoirs," Transactions of the American Society of Civil Engineers, 116, 1951 (the estimator later named the Hurst parameter).

## References — self-similarity and its consequences

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- W. E. Leland, M. S. Taqqu, W. Willinger, D. V. Wilson, "On the Self-Similar Nature of Ethernet Traffic (Extended Version)," IEEE/ACM Transactions on Networking, 2(1), 1–15, 1994.
- W. Willinger, M. S. Taqqu, R. Sherman, D. V. Wilson, "Self-Similarity Through High-Variability: Statistical Analysis of Ethernet LAN Traffic at the Source Level," IEEE/ACM Transactions on Networking, 5(1), 71–86, 1997.
- K. Park, W. Willinger, "Self-Similar Network Traffic: An Overview," in *Self-Similar Network Traffic and Performance Evaluation*, Wiley, 2000.
- V. Paxson, S. Floyd, "Wide Area Traffic: The Failure of Poisson Modeling," IEEE/ACM Transactions on Networking, 3(3), 226–244, 1995.

# Thank you for your attention

Jan Rozhon, Miroslav Vozňák

+420 596 995 900 · [jan.rozhon@vsb.cz](mailto:jan.rozhon@vsb.cz)

[www.vsb.cz](http://www.vsb.cz)



# Extras

Estimating the Hurst parameter, and a model complexity ladder

# Extras — estimating H from a finite trace

H is a parameter of an infinite series; every practical method **estimates** it from a finite measured trace.

## Variance-time plot

Aggregate the trace at increasing block sizes  $m$  and plot  $\log(\text{Var}(X^{(m)}))$  against  $\log m$ . Poisson-like traffic decays with slope  $-1$ ; self-similar traffic decays more slowly, with slope  $2H - 2$ .

## Rescaled range (R/S) statistic

Plot  $\log(R/S)$  against  $\log n$  for windows of length  $n$ . The slope of the resulting line is a direct estimate of H — Hurst's original 1951 method, from reservoir data.

Both are simple to implement but biased on short traces; wavelet-based estimators are preferred in modern traffic-analysis tooling.

## Extras — the loss-model complexity ladder

| Model           | Parameters                                       | What the new parameter buys                                         |
|-----------------|--------------------------------------------------|---------------------------------------------------------------------|
| Bernoulli       | $p$                                              | baseline — independent loss, no memory                              |
| Simple Gilbert  | $p, q$                                           | state memory $\rightarrow$ loss bursts                              |
| Gilbert         | $p, q, h$                                        | probabilistic recovery <i>within</i> a burst                        |
| Gilbert-Elliott | $p, q, h, k$                                     | probabilistic loss <i>within</i> a good period too                  |
| Four-state      | $p_{41}, p_{14}, p_{13}, p_{31}, p_{32}, p_{23}$ | independently shaped burst-length and good-run-length distributions |

Fitting more parameters needs more data and a longer trace before the estimate is trustworthy — pick the simplest model that reproduces the burst statistics you actually measured, not the most elaborate one available.